Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Songhak Yoon	Email	song.hak.yoon@iwks.fraunhofer.de
Institute Name	Fraunhofer IWKS			
Address	Aschaffenburger Str. 121, 63457 Hanau, Germany			

2. Technology Overview

Title of Proposed Technology	**Development of a process technology for NFM-based layered cathode materials for sodium-ion batteries using Taylor flow**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Chemical Processes	*Process Equipment Technologies/ Process System Technologies*	*Batteries*	*Battery Materials/ Secondary Batteries*
Concept of Proposed Technology	1) The necessity for research and development into cathode materials for sodium-ion batteries - Sodium-ion batteries are rapidly emerging as next-generation secondary battery technology due to their raw material supply stability, cost competitiveness, and suitability for large-scale energy storage. Consequently, overcoming the energy density limitations of cathode materials has become a key challenge for technological competitiveness. NASICON and polyanion-based cathode materials exhibit excellent structural stability and longevity characteristics. However, their relatively low operating voltage and limited theoretical capacity impose structural limitations on scaling up for medium-to-large-scale ESS and high-power/high-energy applications. - layered transition metal oxide cathode materials, particularly the NFM (Na–Fe–Mn–O) series of			

multicomponent layered cathodes, are gaining attention as a promising alternative capable of substantially overcoming the energy density limitations of sodium-ion batteries. NFM-series materials can simultaneously achieve relatively high operating voltages and excellent reversible capacity. Through transition metal composition design, they offer high freedom in adjusting electrochemical properties, positioning them as key candidates for realizing next-generation high-energy sodium-ion batteries.

- In conventional batch synthesis processes, composition uniformity, particle size and microstructure control, and crystalline phase stability are highly dependent on process conditions. Low process reproducibility leads to a recurring issue where quality deviations structurally amplify during scale-up. This is judged to stem not from inherent limitations of NFM-series materials, but rather from the insufficient establishment of process technologies suitable for high-energy layered cathode materials.

- To achieve stable characteristics and industrial application of NFM-series layered cathode materials, it is essential to develop a continuous synthesis process technology capable of precisely controlling composition, reaction conditions, and microstructure, alongside a manufacturing system with assured process reproducibility. This is regarded as a key technological challenge for overcoming the structural limitations of existing batch processes and realizing the mass production and commercialization of sodium-ion battery cathode materials.

2) The necessity of developing NFM-based cathode synthesis process technology

- The synthesis of NFM-series layered cathode materials for sodium-ion batteries has been sufficiently validated in literature and at the laboratory level. However, there are clear limitations to directly applying these synthesis methods on an industrial scale; these limitations stem not from the synthesis recipe itself, but from constraints within the process structure.

a) Limitations of precise control in batch processes

Batch synthesis processes struggle to precisely control the interrelationship between composition, reaction conditions, and microstructure. For NFM-series materials, phase stability and electrochemical performance vary significantly depending on metal ion composition, sodium vacancies, and reaction environment. In batch processes, the mixing, reaction, precipitation, and crystal growth stages proceed non-uniformly over time, reducing the reproducibility of the final product.

b) Amplification of Quality Deviations During Scale-up

Minor variations in process conditions tolerable at laboratory scale become amplified during mass production, manifesting as particle size distribution, phase impurities, and compositional deviations. This directly leads to unstable cathode performance, and fundamental resolution is difficult to achieve solely through adjusting certain synthesis conditions.

(max 10 pages)

	c) Strong Interdependence Between Reaction Stages NFM-series materials exhibit strong interdependencies between reaction rate, nucleation, and crystal growth stages, necessitating integrated control of the entire process as a single system. Partial optimization of individual stages has limitations, requiring technology capable of precisely controlling reaction conditions within a continuous process flow.
Rationale for Joint R&D Project	- This joint R&D project was conceived to substantially enhance the industrial readiness of NFM-series layered cathode materials for sodium-ion batteries. Whilst the synthesis mechanism and material properties of NFM-series cathodes have been sufficiently validated at the laboratory level, technical gaps remain in converting them into a process structure capable of continuous operation and achieving stable implementation in an industrial setting. - Fraunhofer, as a research institution possessing experience in pilot-scale process operation and process scaling verification, will perform the role of systematically analyzing and verifying the process stability, operational limits, and technical risks during the scaling process of the synthesis system converted to a continuous process. This will objectively evaluate whether the synthesis conditions secured in the laboratory can be maintained during scale-up and verify industrial applicability. - Korean SMEs, as the technology holders having secured the synthesis mechanism and laboratory-level synthesis conditions for NFM-based cathode materials, lead the design of a continuous synthesis process based on Taylor fluid flow and the transition of the process structure. They implement conditions established in the existing batch process into a structure capable of continuous operation, and by accumulating process control technology and operational know-how, they drive process advancement centered on industrial sites. - This joint R&D aims to advance process technology linking NFM synthesis technology to industrial application. By synergistically combining Fraunhofer's process scaling verification capabilities with Korean SMEs' continuous process development expertise, it simultaneously ensures material performance reproducibility, process stability, and long-term operational reliability. This collaborative model organically links research, process development, and industrial application, and is considered an essential structure for enhancing the practical commercialization potential of NFM-based sodium-ion battery cathode materials.
Key Words	1. Cathode materials for sodium-ion batteries 2. NFM-series layered oxides 3. Material design and synthesis technology 4. Continuous synthesis process using Taylor-type fluid flow 5. Industrial application and process scalability

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3 - 4)	End of the Project	Level (5 - 6)
R&D Objectives	The ultimate objective of this research and development project is to transition the NFM-series sodium-ion battery layered cathode material synthesis technology secured by Korean SMEs at the laboratory and pre-production stages and to combine this with Fraunhofer Institute's process scale-up and pilot verification capabilities. This aims to advance NFM-series high-energy cathode materials, previously confined to the research level, into practical industrial application-ready process technologies. Therefore, this project is not merely research to improve material performance; its core objective is to structurally demonstrate the industrial viability of NFM-series cathodes through a continuous process-based approach. 1) Establishment of continuous synthesis process technology for NFM-series cathode materials This project aims to overcome the structural limitations of existing batch synthesis processes and establish a synthesis technology with inherent process reproducibility. This will be achieved by systematically applying a continuous process based on Taylor fluid flow to the synthesis of NFM-series layered cathode materials. The objective is to minimize batch-to-batch quality variation and enhance process reproducibility through precise control of metal ion composition, reaction conditions, residence time, and mixing environment. 2) Development of manufacturing technology for cathode materials with assured compositional uniformity and crystalline phase stability Layered cathode materials of the NFM series are process-sensitive materials where crystalline phase stability and electrochemical performance are significantly influenced by composition and Na vacancy changes. This project aims to develop manufacturing technology that simultaneously ensures compositional uniformity and crystalline phase stability throughout the entire synthesis process by leveraging the advantages of continuous processing. 3) Securing process stabilization technology enabling prolonged continuous operation This project aims to secure process stabilization technology that minimizes fluctuations in particle characteristics, composition, and crystalline phase even under the prolonged continuous operation conditions required in industrial environments, going beyond short-term synthesis verification at laboratory scale. Furthermore, by leveraging Fraunhofer's process scaling and pilot verification capabilities to demonstrate the long-term operational stability and scalability of the continuous synthesis process, it seeks to clearly validate the industrial applicability of NFM-series high-energy cathode materials.			
R&D Target Specifications	The objective of this research and development is to establish a continuous synthesis process system capable of industrially producing NFM-series layered cathode materials for sodium-ion batteries. To this end, the key performance targets are set as improving production throughput,			

	ensuring process stability, enabling long-term continuous operation, and securing batch-to-batch quality reproducibility. - This project focuses on establishing a continuous synthesis system based on Taylor flow and securing technology for the precise control of reaction conditions and key process variables. The developed process system will not replace existing synthesis technologies but will function as a process and system technology that enhances the industrial applicability of already validated material synthesis techniques. - It is anticipated that this will establish the foundational technology for the mass production and commercialization of cathode materials for sodium-ion batteries.
R&D Plan	This research adopts as its core strategy the transition from the existing batch synthesis-centered material development paradigm to an industrial-oriented R&D approach based on continuous processes. It combines the synthesis mechanism and laboratory-scale optimal conditions for NFM-series layered cathode materials accumulated by Korean SMEs with Fraunhofer's specialized capabilities in process scaling and pilot operation. This approach converts existing batch processes into structures capable of continuous operation, progressively verifying quality reproducibility and process stability during the scale-up process. The core research tasks of this project are as follows: 1. Design of a continuous synthesis process for NFM-series layered cathode materials 2. Securing precision control technology for key process variables such as reaction conditions, residence time, and mixing environment 3. Development of process stabilization and clogging/deposition suppression technologies enabling prolonged continuous operation 4. System optimization to ensure quality reproducibility and process yield Regarding research methodology, continuous reactor-based synthesis experiments will be conducted in stages. Quantitative analysis will assess composition uniformity, particle size distribution, crystalline phase stability, and process operational stability in response to variations in key process variables such as flow rate, concentration, reaction time, and temperature. Crucially, this will not be limited to evaluating performance based on short-term synthesis results. Instead, comprehensive assessment will be conducted through extended continuous operation trials, evaluating process interruption occurrence, quality variability, and yield changes. This will objectively validate the practical applicability of the continuous process. Based on these results, a cyclical R&D strategy will be applied, involving iterative feedback and improvement of process conditions. The phased milestones are set as follows: Stage 1: Precisely analyze the limitations of the existing batch process and design a continuous

	process structure reflecting these insights to establish fundamental process conditions enabling short-term continuous operation. Stage 2: Verify quality reproducibility and process stability through optimization of key process variables and extended continuous operation testing. Stage 3: Substantially enhance the maturity of the process technology for industrial application by advancing the continuous process system and verifying process scalability. The research schedule is structured to focus initially on process design and short-term operation verification, gradually expanding to extended continuous operation and production volume verification stages. The ultimate goal by the research completion date is to establish a continuous operation-capable synthesis process system for NFM-series high-energy sodium-ion battery cathode materials, designed for industrial application.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Fraunhofer will be responsible for scaling up the continuous synthesis process to pilot scale and conducting technical validation in this joint R&D project. This will be based on the synthesis mechanism and laboratory-level synthesis conditions for NFM-series sodium-ion battery cathode materials secured by the Korean SME. The focus will be on the role of a specialist research institute analyzing and validating technical issues that may arise during the process scaling-up phase, rather than solely providing the core synthesis technology. - Fraunhofer will systematically evaluate process stability, operational limits, changes in heat and mass transfer characteristics, and factors affecting quality reproducibility by applying long-term continuous operation conditions to the continuous synthesis system converted to a Taylor flow-based configuration. It will quantitatively analyze material property variations resulting from changes in process conditions, particularly focusing on whether the synthesis conditions established in the laboratory can be maintained during scale-up. - Furthermore, through pilot-scale process operation tests, Fraunhofer verifies the scalability and operational stability of the continuous process. Changes in particle size distribution, composition uniformity, and crystal structure stability are confirmed via XRD analysis, composition analysis, and physical property evaluation. These results serve as a basis for pre-emptively identifying potential technical risks that may arise during future industrial application stages. - Through this, Fraunhofer enhances the completeness and reliability of continuous synthesis process technology, playing a pivotal role in assessing technical feasibility and managing risks from a process scaling perspective.	

(max 10 pages)

Expected R&R of Korean SME Partner	- Korean SMEs participate as key contributors in developing a continuous synthesis process utilizing Taylor fluid flow, based on the synthesis mechanism of NFM-series sodium-ion battery cathode materials. They lead the design of continuous reactors, the setting of key process variables, and the establishment of process control strategies to convert the existing batch process into a structure capable of continuous operation. - During the research process, quality indicators such as particle size distribution, composition deviation, and crystalline structure stability are evaluated through short- and long-term continuous operation tests. Design and operational criteria are established to minimize operational risks such as clogging and precipitation. Furthermore, electrochemical and structural analyses are conducted in collaboration with domestic universities, with findings incorporated into process optimization and stability assurance. - The derived continuous process technology is validated at pilot scale through collaboration with Fraunhofer. Korean SMEs will then lead the implementation of industrially applicable continuous process technology based on this. This strengthens process development capabilities and secures a foundation for future expansion into production processes.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this joint R&D project are to develop a continuous synthesis process system for NFM-series layered cathode materials for sodium-ion batteries and to establish the foundation for connecting this to industrial production. This project focuses on advancing the technology to a production process system suitable for industrial application, based on the synthesis technology for materials with uniform characteristics produced continuously. Consequently, both technological achievements and industrial ripple effects are anticipated simultaneously. 1) Technical Outcomes First, the implementation of a continuous synthesis process system using Taylor fluid flow and the establishment of its operating conditions are anticipated as key technical outcomes. Korean SMEs will design and construct a continuous process based on the synthesis mechanism of NFM-series cathode materials and synthesis conditions established at the laboratory scale. They will then validate the process performance against the quantitative targets set for this project (continuous operation time ≥ 3 days, process yield ≥ 90%). This will establish a technological foundation to secure continuous operation stability required in industrial environments, moving beyond existing short-term synthesis at the laboratory level.

Second, a database correlating process conditions and material properties will be constructed. Fraunhofer will quantitatively analyze the effects of residence time, mixing intensity, and reaction condition variations on composition uniformity, crystal structure stability, and particle size characteristics through pilot-scale process operation tests. Based on this, it will preemptively identify potential technical risks during continuous process scaling. This data will serve as a technical foundation applicable to future development of similar compositions or scaled-up NFM-series cathode materials.

2) Commercialization Potential

The continuous synthesis process system derived from this project aims to achieve a level of technological maturity sufficient for pilot-scale process validation for industrial application, rather than directly performing commercial mass production. By verifying process feasibility and operational stability during the research phase and establishing the process condition tolerance range (window) and quality control (QC) criteria, the potential for future process scaling required in an industrial environment can be systematically evaluated.

Furthermore, this technology possesses the potential for expansion and application to various NFM series compositions and next-generation sodium-ion battery cathode materials, leveraging the structural advantages inherent in continuous processes. In other words, it is not limited to a single composition but holds significant commercialization value from a technological perspective, as it possesses the potential to evolve into a continuous process-based cathode material production process platform technology.

3) Industrial/Economic Impact

First, this project contributes to securing low-cost, mass-applicable process technologies demanded by the sodium-ion battery market. Particularly in the rapidly growing sodium-ion battery market, centered on ESS and mid-to-low-end electrification sectors, process stability and quality reproducibility act as core competitive factors alongside material performance itself. Securing continuous process technology provides a crucial foundation for meeting these market demands.

Second, from the perspective of Korean SMEs, this project enables the internalization of process system technology and equipment-based technical capabilities. This extends beyond simple material supply to encompass the capability to provide integrated solutions, including process design and operation technologies. This can lead to securing technological competitiveness in future domestic and international battery material and equipment markets.

Third, combining Fraunhofer's process validation capabilities with Korean SMEs' process

	implementation capabilities can minimize potential technical risks during process scaling. This project thus bridges the technological gap between the research stage and industrial application, contributing not only to short-term outcomes but also to long-term supply chain stability and the formation of a technology cooperation framework. 4) Overall Expected Effects Overall, this joint R&D project advances the continuous synthesis process system for NFM-based sodium-ion battery cathode materials to a level verifiable for industrial application. This provides a technological foundation scalable to future commercial production stages. It is expected to enhance the commercialization potential of sodium-ion battery cathode materials and contribute to strengthening the competitiveness of related industries.

6. References

(1) International Energy Agency (IEA), Global EV Outlook 2025, IEA Publications, Paris, France, 2025.

(2) Fraunhofer Institute for Systems and Innovation Research (ISI), Lithium-Ion Battery Roadmap: Industrialization Perspectives Toward 2030, Karlsruhe, Germany, 2023.

(3) Fraunhofer Institute for Systems and Innovation Research (ISI), Alternative Battery Technologies Roadmap 2030+, Karlsruhe, Germany, 2022.

(4) International Renewable Energy Agency (IRENA), Sodium-ion Batteries: A Technology Brief, Abu Dhabi, UAE, 2025.

(5) Battery 2030+ Initiative, BATTERY 2030+ Roadmap, European Union, 2022.

(6) Delmas, C., Braconnier, J. J., Fouassier, C., & Hagenmuller, P., “Electrochemical intercalation of sodium in Na_xCoO_2 bronzes,” Solid State Ionics, vol. 3–4, pp. 165–169, 1981.

(7) Yabuuchi, N., Kubota, K., Dahbi, M., & Komaba, S., “Research development on sodium-ion batteries,” Chemical Reviews, vol. 114, no. 23, pp. 11636–11682, 2014.

(8) Slater, M. D., Kim, D., Lee, E., & Johnson, C. S., “Sodium-ion batteries,” Advanced Functional Materials, vol. 23, no. 8, pp. 947–958, 2013.

(9) Gonzalo, E., Han, M. H., López del Amo, J. M., Acebedo, B., Casas-Cabanas, M., & Rojo, T., “Synthesis and characterization of pure P2- and O3-$Na_{2/3}Fe_{2/3}Mn_{1/3}O_2$ as cathode materials for sodium-ion batteries,” Journal of Materials Chemistry A, vol. 2, pp. 18523–18530, 2014.

(10) Tapia-Ruiz, N., Armstrong, A., Alptekin, H. et al, “2021 roadmap for sodium-ion batteries,” Journal of Physics: Energy, vol. 3, no. 3, 2021.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization